Data-Oriented Programming

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Dr. Michael Sperber

- Geschäftsführer Active Group
- Promotion Universität Tübingen
- funktionale Programmierung
- Softwarearchitektur
- Ausbildung
- programmiere seit 1983
  erstes Buch 1986

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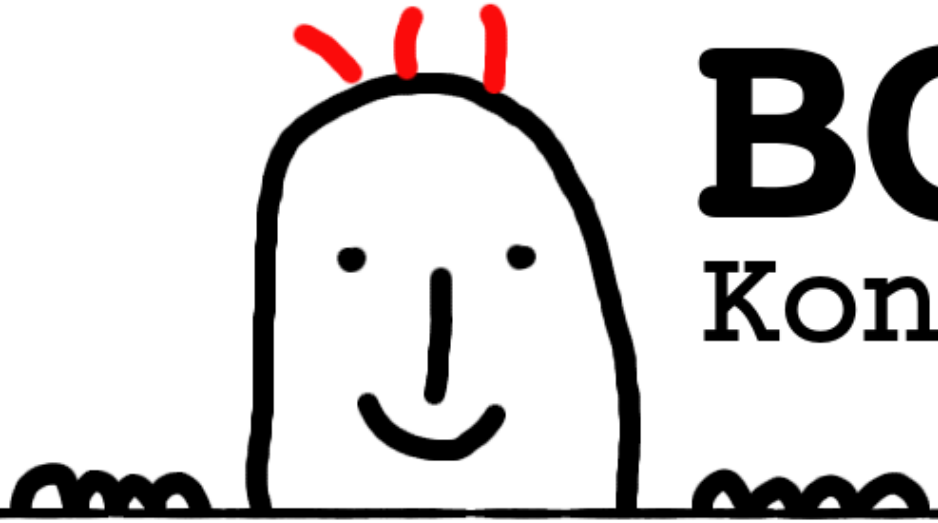
- Individualsoftware
- branchenunabhängig
- funktionale Programmierung
- Schulungen, Coaching
- iSAQB-akkreditiert
  Funktionale Softwarearchitektur
  Domänenspezifische Sprachen
  Formale Methoden

FUNAR 23.3. HH

www.active-group.de

funktionale-programmierung.de



BOB 2026

Konferenz

13 March

bobkonf.de

Berlin

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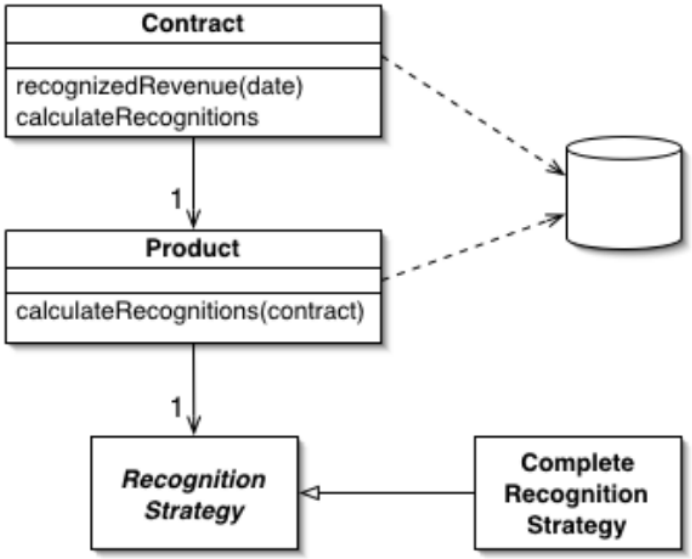
Domain Model

An object model of the domain that incorporates both behavior and data.

For a full description see P of EAA page **116**



At its worst business logic can be very complex. Rules and logic describe many different cases and slants of behavior, and it's this complexity that objects were designed to work with. A Domain Model creates a web of interconnected objects, where each object represents some meaningful individual, whether as large as a corporation or as small as a single line on an order form.

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Objekte und ihr Zustand

Quelle: https://commons.wikimedia.org/wiki/File:Gotisches_Haus_%28Brandenburg_an_der_Havel%29.jpg
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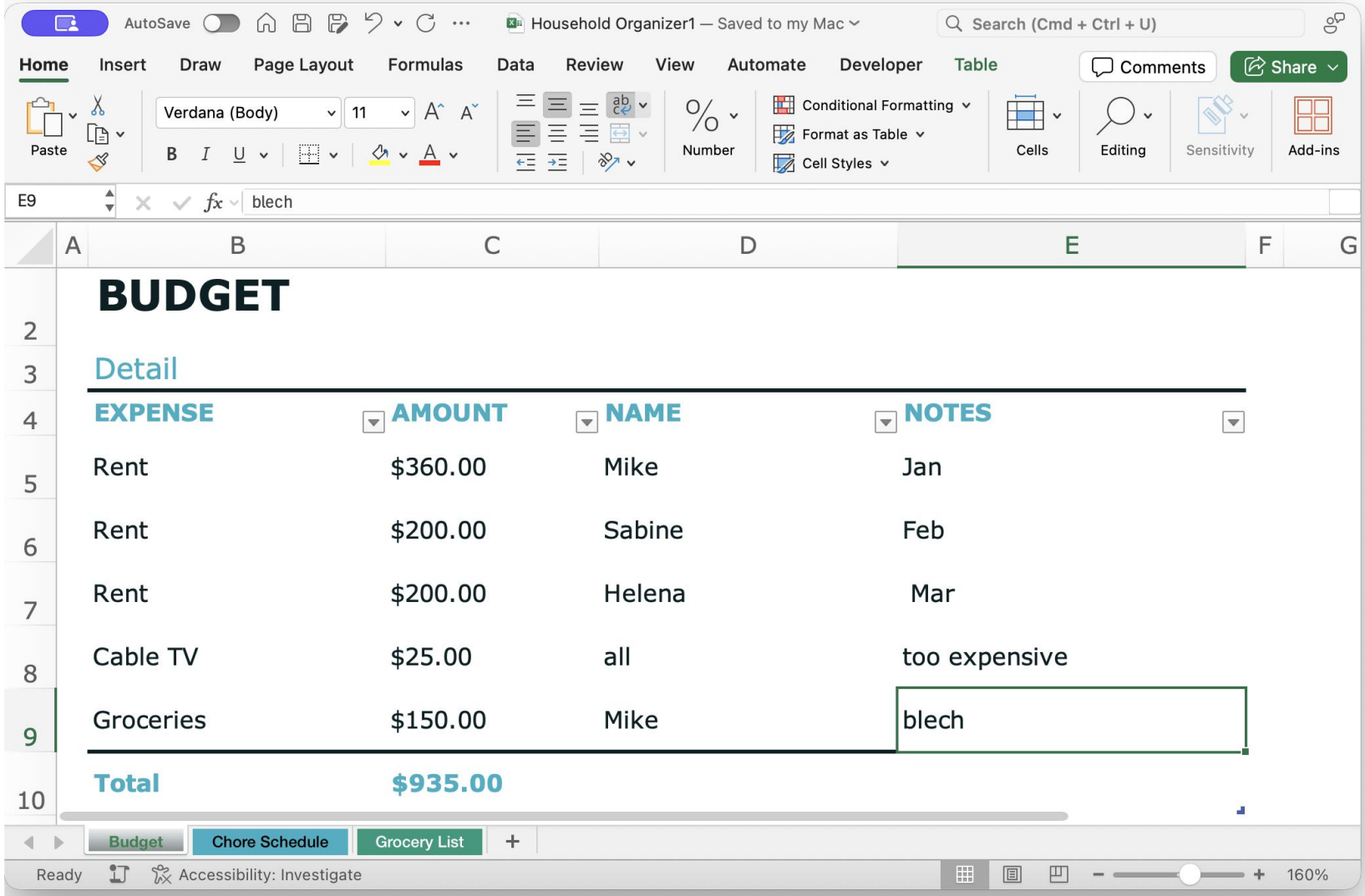
Alan Kay on OO

"Though OOP came from many motivations, two were central. [...] to find a more flexible version of assignment, and then to try to eliminate it altogether."

Alan Kay, *History of Smalltalk*
Communications of the ACM, 1996

DATA UND FUNCTIONS...
EVERYWHERE

Rectangular Design



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Funktionale Programmierung

- (fast) alles unveränderlich
- höherstehende Abstraktionen
- Mathematik FTW
- Kombinatormodelle

Data-Oriented Programming

- algebraische Datentypen
  Summen & Produkte
- Pattern Matching
- unveränderliche Daten

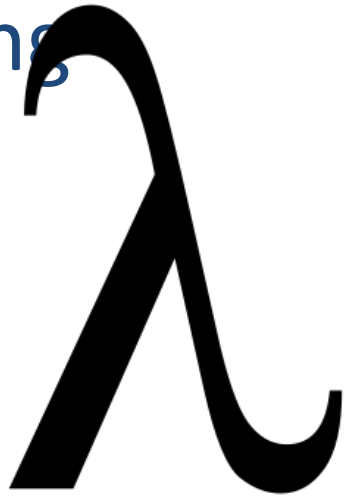
https://youtu.be/1dY57CDxR14?si=ZUIKApDZ94_ebMGg

Funktionale Programmierung

= Data-Oriented Programming

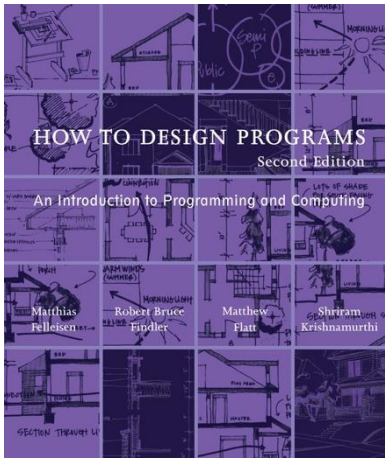
+ Abstraktion

+ (mathematische) Semantik



Konstruktionsanleitungen/ design recipes Data-Orientierte Programmierung/ Funktionale Programmierung

- systematische Design-Methode
- bottom-up
- reichhaltige Datenmodelle
- niedrige Kopplung durch Immutability
- sehr lernbar



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Realität

Id	Art	Name	Kunde	Datum 1	Beitrag	Prop1 (NULL)	Prop2 (NULL)	Prop3 (NULL)	Prop4 (NULL)	Prop5 (NULL)	Prop5 (NULL)	Prop6 (NULL)	Prop7 (NULL)	Prop8 (NULL)

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Repo

https://github.com/active-group/architecture-summit-2026

Closure of Operations

"Where it fits, define an operation whose return type is the same as the type of its argument(s). If the implementer has state that is used in the computation, then the implementer is effectively an argument of the operation, so the argument(s) and return value should be of the same type as the implementer. Such an operation is closed under the set of instances of that type. A closed operation provides a high-level interface without introducing any dependency on other concepts."

Evans, *Domain-Driven Design*

Bühnenbeleuchtung

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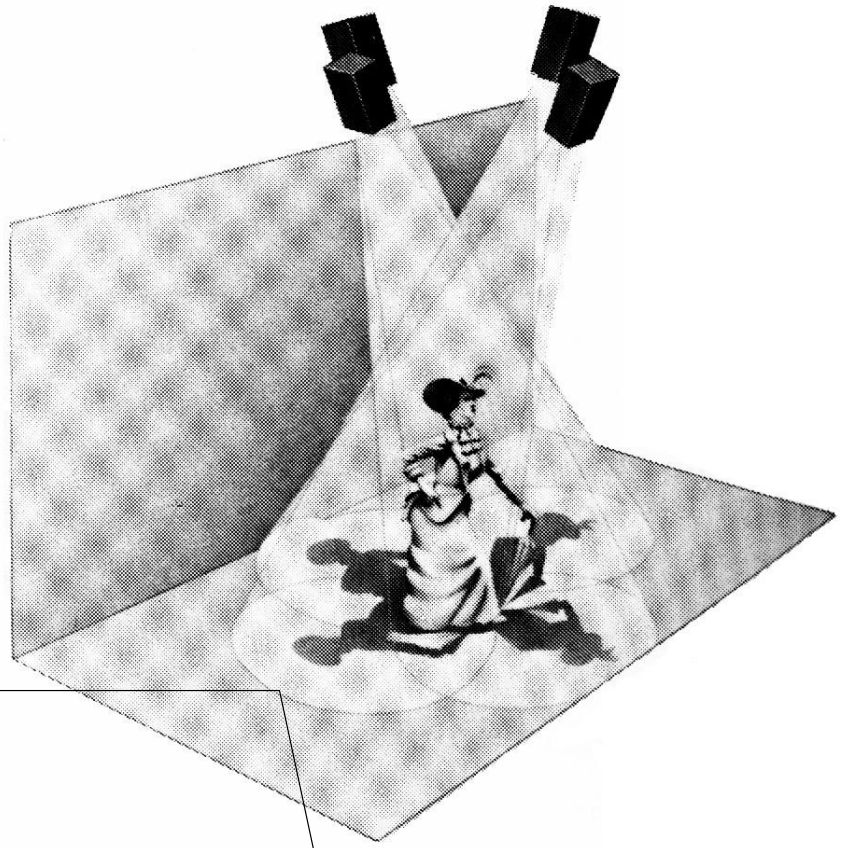
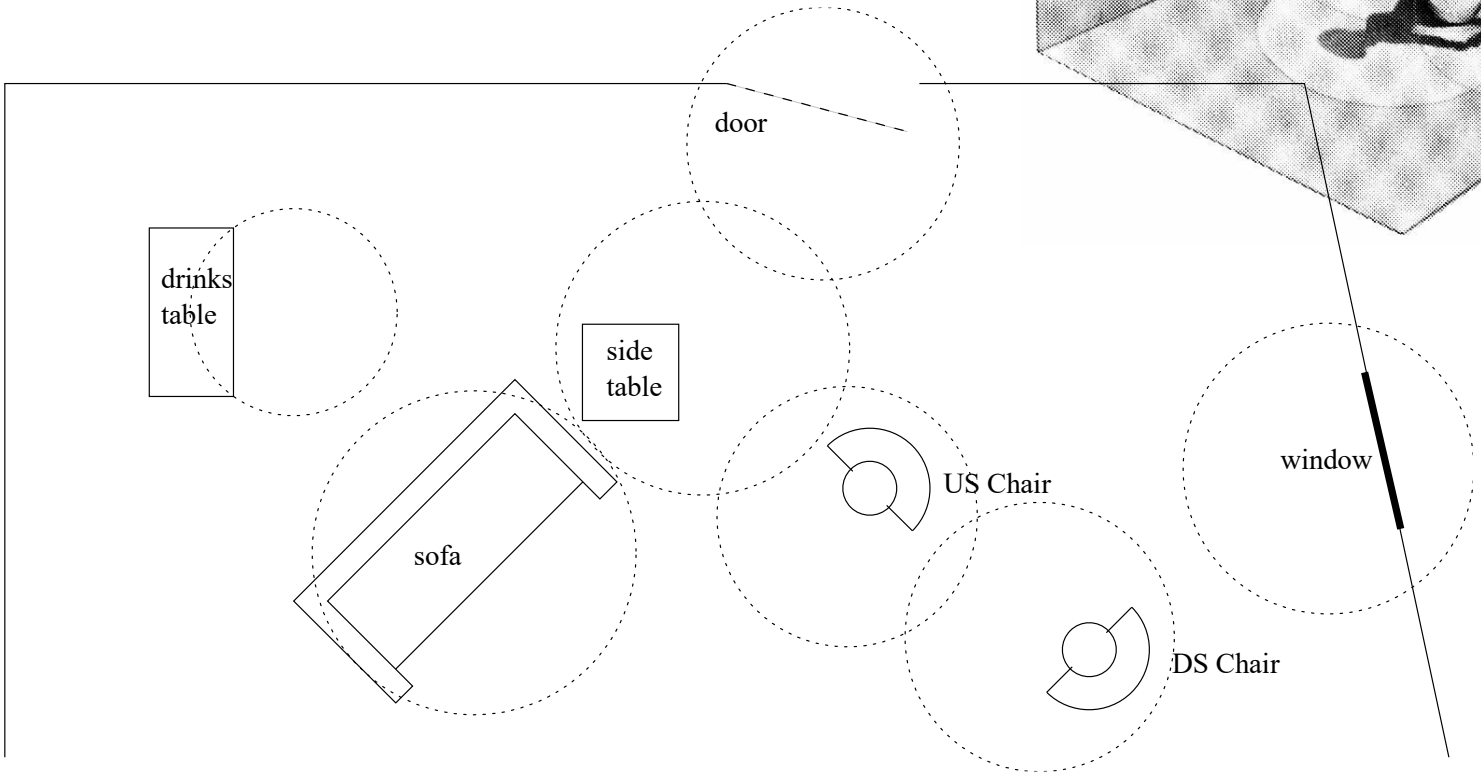
Beleuchtungsdesign





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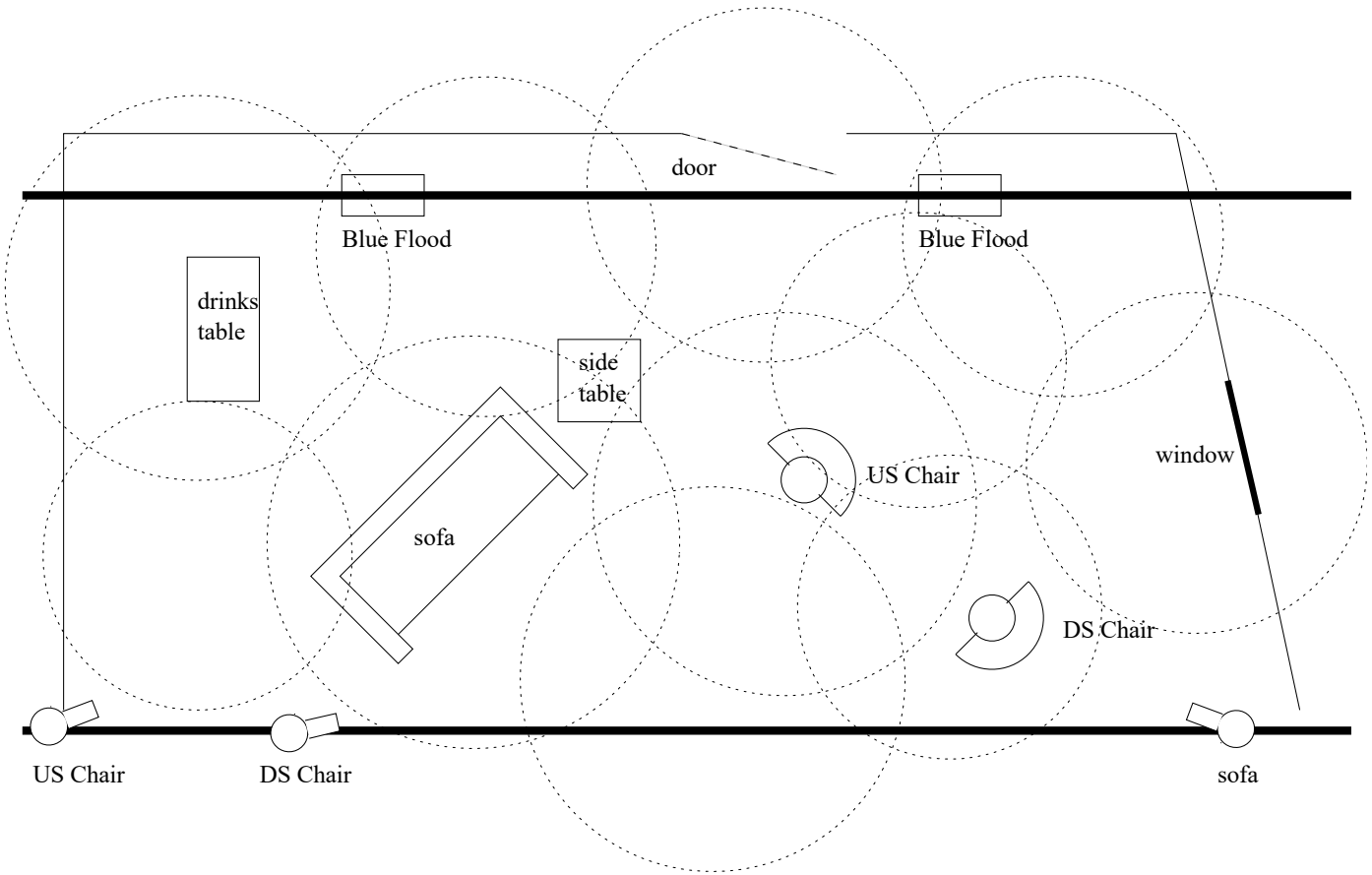
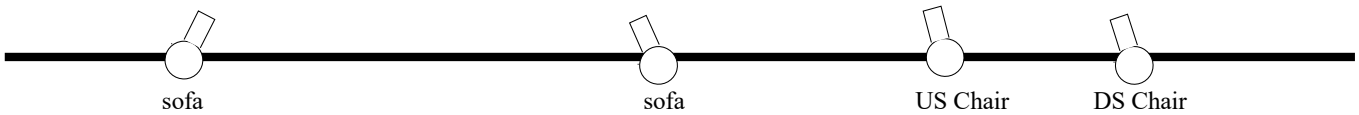
Spezifikation vs. Implementierung





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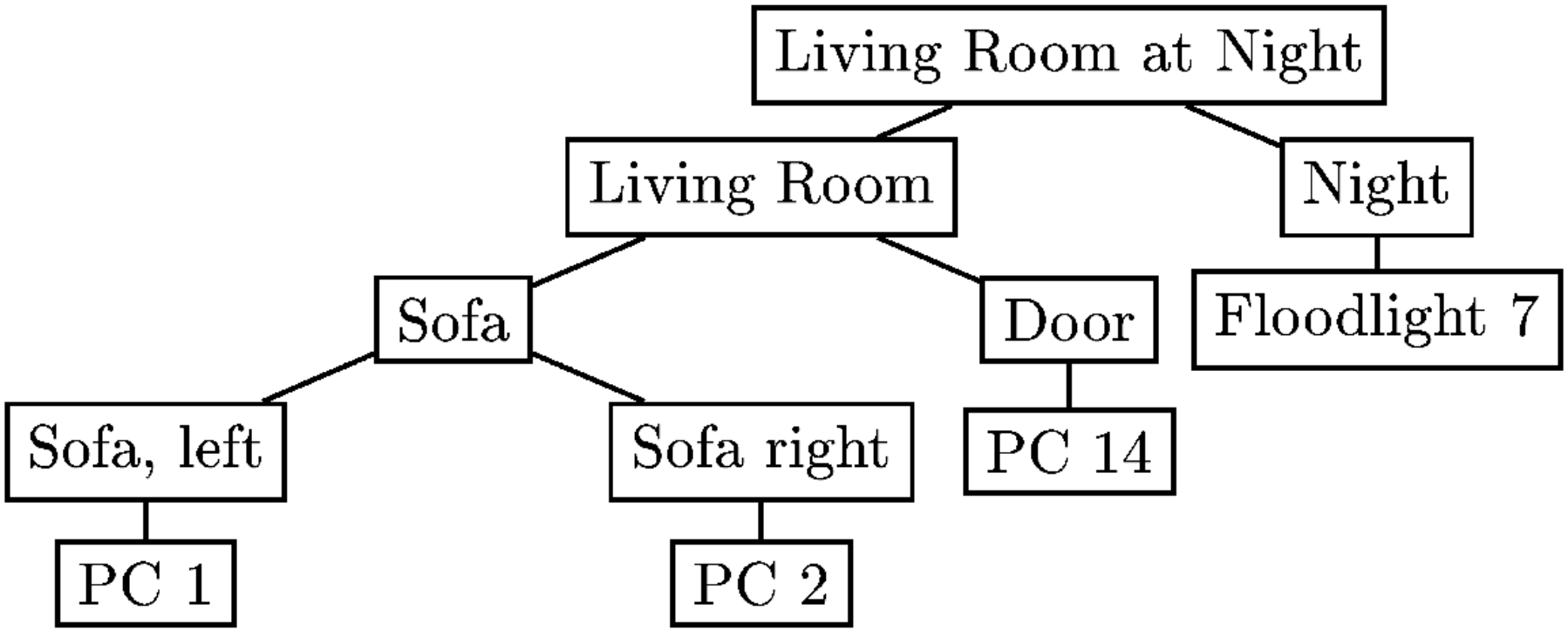
Hierarchische Struktur



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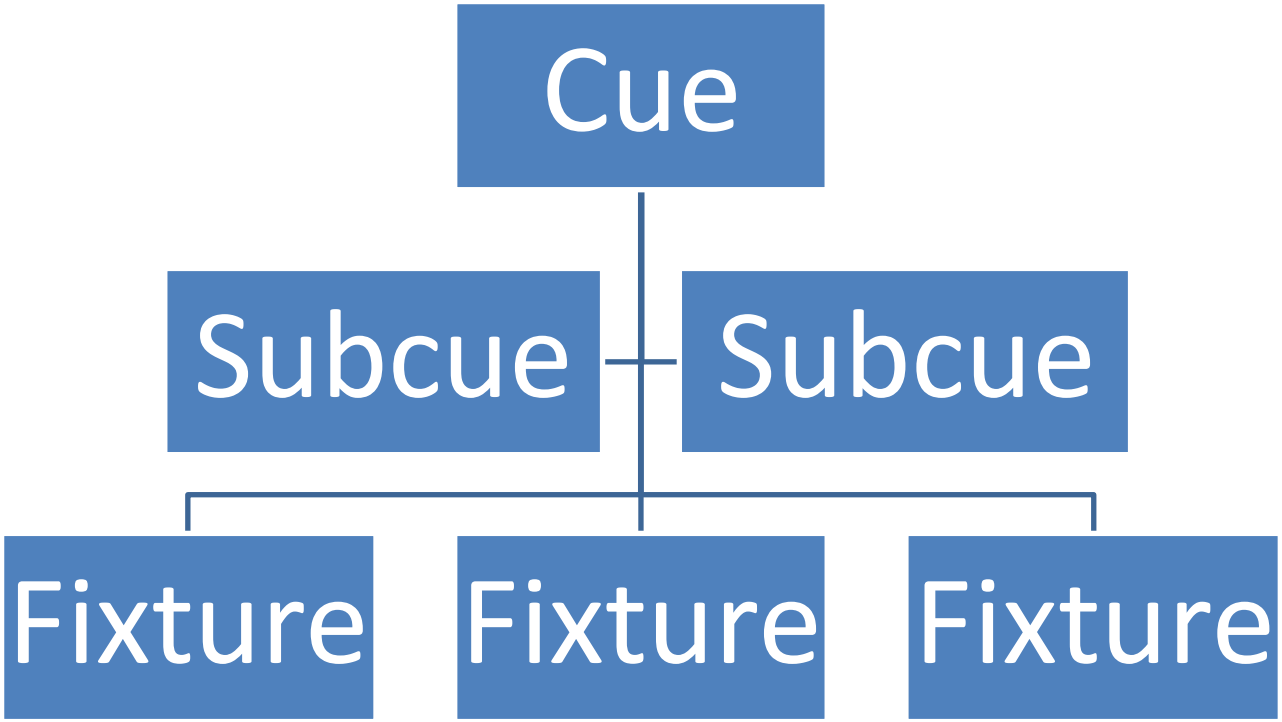
Beleuchtungssteuerungssysteme

- “Submaster”
- “relative Cues”
- “Presets”
- …

Daten-Schichtenmodell



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Kombinatormodell

Ein **Cue** ist ...

- ... eine einzelne **Lampe**
- ... ein **skalierter** Cue
- ... eine **Kombination** von Cues

```
type Number = Int
type Factor = Double

data Cue
  = Fixture Number
  | Scaled Factor Cue
  | Combination [Cue]
```

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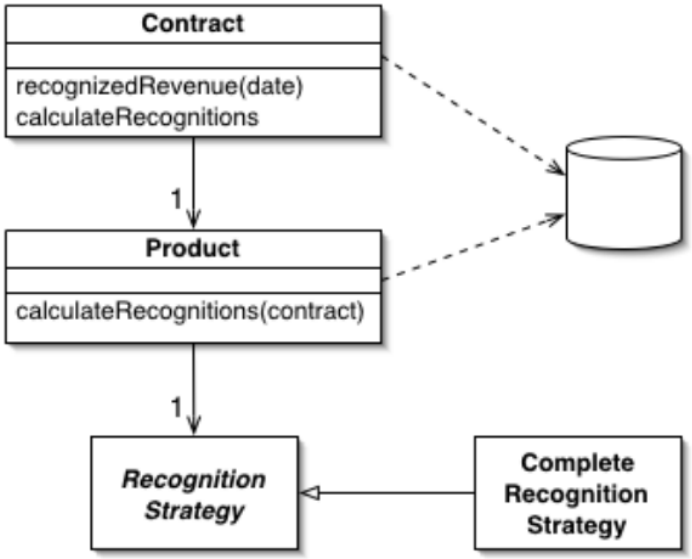
Domain Model

semantics

An object model of the domain that incorporates both ~~behavior~~ and data.

For a full description see P of EAA page **116**



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Denotationelle Semantik

```
type Meaning = Number -> Intensity

denotation :: Cue -> Meaning

denotation (Fixture number) = \ number' ->
  if number == number'
  then 1.0
  else 0.0

denotation (Scaled factor cue) =
  scale factor (denotation cue)

denotation (Combination cues) =
  foldr supremum
      (const 0.0) (map denotation cues)
```

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Principle of compositionality

Article Talk

From Wikipedia, the free encyclopedia

(Redirected from Compositionality)

In semantics, mathematical logic and related disciplines, the **principle of compositionality** is the principle that the meaning of a complex expression is determined by the meanings of its constituent expressions and the rules used to combine them. The principle is also called **Frege's principle**, because Gottlob Frege is widely credited for the first modern formulation of it. However, the principle has never been explicitly stated by Frege,[1] and arguably it was already assumed by George Boole[2] decades before Frege's work.

https://en.wikipedia.org/wiki/Principle_of_compositionality

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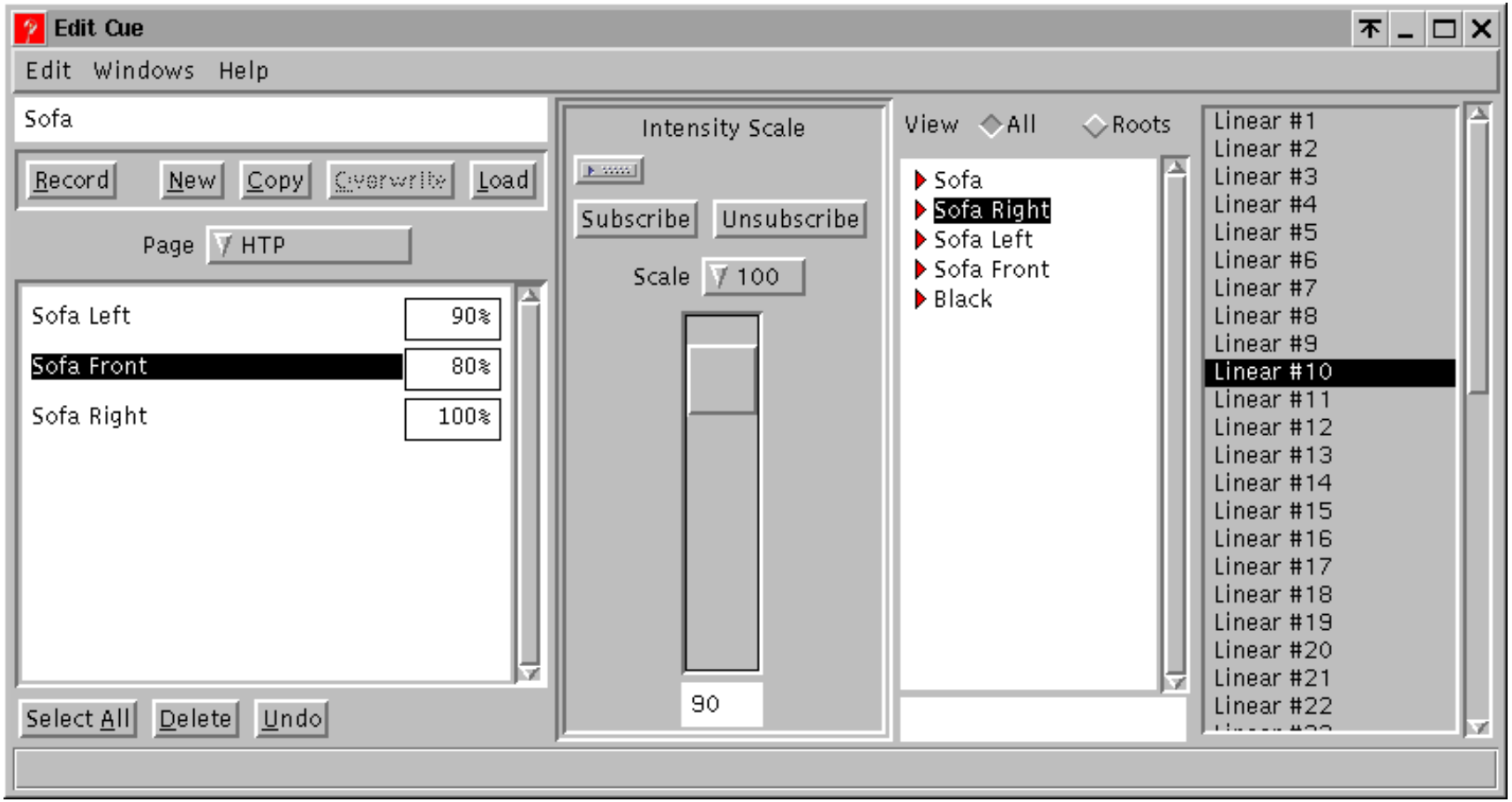
Lula


Edit Cue
Edit Windows Help
Sofa
Record New Copy Overwrite Load
Page HTP
Sofa Left 90%
Sofa Front 80%
Sofa Right 100%
Select All Delete Undo
Intensity Scale
Subscribe Unsubscribe
Scale 100
90
View All Roots
Sofa
Sofa Right
Sofa Left
Sofa Front
Black
Linear #1
Linear #2
Linear #3
Linear #4
Linear #5
Linear #6
Linear #7
Linear #8
Linear #9
Linear #10
Linear #11
Linear #12
Linear #13
Linear #14
Linear #15
Linear #16
Linear #17
Linear #18
Linear #19
Linear #20
Linear #21
Linear #22

Vorteile

- **viel** einfacher zu benutzen
- **viel** schneller zu benutzen
- braucht keine große Konsole
- zieht Änderungen mit
- funktioniert auf Tournee
- langlebig